Chiba University
Graduate School of Advanced Integration Science

Master's Thesis

Quantum algorithm for finding all eigenvalues of Hermitian matrix

エルミート行列のすべての固有値を探す量子アルゴリズム

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Chapter.1

INTRODUCTION

1.1 BACKGROUND

Quantum computing is one of the important technologies in the future and uses the property of quantum mechanics, the most famous example being superposition. It makes quantum algorithms such as Shor's factoring algorithm, Grover's algorithm, quantum phase estimation and Quantum algorithm for linear systems of equations perform exponential speed ups over their classical counterparts, and has the potential to be further developed to use in wide range of field such as chemistry, AI, operations research and simulation. One of the most famous examples is to use quantum computers for simulating other quantum systems.

Eigenvalues and Eigenvectors playing an important role in virtually all field of science and engineering have many important applications in computer vision and machine learning. Well known examples are PCA(Principal Component Analysis) for dimensionality reduction and Eigenfaces for face recognition. In quantum chemistry, one of the most important work is finding eigenvalue of Hamiltonian. Being Hamiltonian's eigenstates corresponding to eigenvalue imply stationary state having a definite energy value and these are the 'energy levels' of a system such as an atom. For a classical computer even to find all eigenvalues of $N \times N$ Hermitian matrix H, these requires time that scales at least as N^3[1] in general.

In recently, to find eigenvalues of Hamiltonian, many quantum algorithms and hybrid quantum-classical algorithms have been developed[2, 3, 4, 5]. In quantum algorithms, precedent algorithms need priori knowledge of eigenvectors and involves work such as using a classical algorithm for initial eigenvectors in advance. Hybrid quantum-classical algorithms are algorithm using quantum algorithm for subroutine in classical algorithm. However, this algorithm has limitation that bottlenecks occur in the part of the classical algorithm.

In this thesis, we focus on the problem of finding eigenvalues of a given matrix. Given an $N \times N$ sparse Hermitian matrix H, in quantum computing, while there are known faster quantum algorithms for finding eigenvalues, all such quantum algorithms require eigenvectors as input. Here, we give a quantum algorithm for finding all eigenvalues of a given Hermitian matrix that do not assume any a priori knowledge of eigenvectors and runs faster than any known classical algorithms on average. The expected time-complexity of

our quantum algorithm is $O(N(\log N)^2)$. Our algorithm is based on random sampling and finds the eigenvalues by combining several fundamental tools in quantum computing, such as Hamiltonian simulations, quantum phase estimation, quantum Fourier transform, and phase kickback.

In remainder of the paper is organized as follows. In section 2, such background knowledge as the back ground and properties of quantum mechanics, and the tool realizing our quantum algorithm is introduced. This section is like a preparation in order to understand the paper. In section 3, the precedent study for obtaining eigenvalues is explained. In section 4, the problem is solved by the method and background knowledge. Finally, Section 5 is contents that have conclusions and a future research.

Chapter.2

PREPARATION

2.1 Properties of Quantum mechanics

Before we describe our algorithm, we explain the properties of quantum and the basic tools and techniques used for quantum computing.

2.1.1 Qubit

The bit is the fundamental concept of classical computation and classical information. Quantum computation and quantum information are built upon an analogous concept, the quantum bit, or shortly *qubit*[6]. A classical bit has an one state such as 0 or 1. However, a qubit can not have only an one state but also multiple states.

2.1.2 Superposition

Superposition is a fundamental principal of quantum mechanics and refers to the quantum phenomenon where a quantum system can exist in multiple states or places at the exact same time[6]. For example, suppose that there are two possible states for a qubit $|0\rangle$ and $|1\rangle$, which as you might guess correspond to the states 0 and 1 for a classical bit. The difference between bits and qubits is that a qubit can be in other state than $|0\rangle$ or $|1\rangle$. It makes possible to form linear combinations of state

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,, \tag{2.1}$$

where the number α and β are complex numbers having condition such as $\sqrt{\alpha^2 + \beta^2} = 1$. The phenomenon is called *Superposition* and coefficient of each state $|0\rangle$ or $|1\rangle$ is related to the probability measured each vector. For example, suppose we measure the state $|\psi\rangle$ in computational state, and then we observe the state $|0\rangle$ or $|1\rangle$. $|\alpha|^2$ means the probability obtaining the state $|0\rangle$ and $|\beta|^2$ means the probability obtaining the other.

2.2 Quantum Fourier transform

Calculation of some transformation on quantum computers can be computed much faster than on classical computers. That has enabled the construction of fast algorithms in quantum computers. One of the transformation algorithms is the discrete Fourier transform. In the usual mathematical notation, the discrete Fourier transform takes as input a vector of complex numbers, $x_0, \cdots, x_{N-1}$ where the length N of the vector is a fixed parameter. It outputs the transformed data such as a vector of complex numbers $y_0, \cdots, y_{N-1}$, defined by

$$y_k \equiv \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j \mathrm{e}^{2\pi i j k/N}. \tag{2.2}$$

There is no difference between the quantum Fourier transform and the conventional discrete Fourier transform except for some conventional notation for quantum Fourier transform[6]. The quantum Fourier transform on an orthonormal basis $|0\rangle, \cdots, |N-1\rangle$ is defined to be a linear operator with the following change the basis states,

$$|j\rangle \rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \mathrm{e}^{2\pi i j k/N} |k\rangle . \tag{2.3}$$

Equivalently, the change an arbitrary state may be written

$$\sum_{j=0}^{N-1} x_j |j\rangle \rightarrow \sum_{k=0}^{N-1} y_k |k\rangle , \tag{2.4}$$

where the amplitudes y_k are the discrete Fourier transform of the amplitudes x_j. Let the basis $|0\rangle, \cdots, |2^n - 1\rangle$ be the computational basis for an n qubit quantum computer with $\mathrm{N} = 2^n$, and the state $|j\rangle$ using the binary represented$j = j_1 j_2 \cdots j_n$. More formally, $j = j_1 2^{n-1} + \cdots + j_n 2^0$. With little algebra the quantum Fourier transform can be given the following useful product representation:

$$|j_1, ..., j_n\rangle \rightarrow \frac{(|0\rangle + \mathrm{e}^{2\pi i 0.j_n} |1\rangle)(|0\rangle + \mathrm{e}^{2\pi i 0.j_{n-1} j_n} |1\rangle) \cdots (|0\rangle + \mathrm{e}^{2\pi i 0.j_1 j_2 \cdots j_n} |1\rangle)}{2^{n/2}}. \tag{2.5}$$

It is so useful that you may even wish to consider this to be the definition of the quantum Fourier transform. The equivalence of the product representation and the definition follows from some elementary algebra:

$$
\begin{aligned}
|j\rangle &\rightarrow \frac{1}{2^{n/2}} \sum_{k=0}^{2^n-1} \mathrm{e}^{2\pi ijk/2^n} |k\rangle && (2.6)\\
&= \frac{1}{2\mathrm{n}/2} \sum_{k_1=0}^{1} \cdots \sum_{k_n=0}^{1} \mathrm{e}^{2\pi ij\left(\sum_{l=1}^{n} k_l 2^{-l}\right)} |k_1 \ldots k_n\rangle && (2.7)\\
&= \frac{1}{2^{n/2}} \sum_{k_1=0}^{1} \cdots \sum_{k_n=0}^{1} \otimes_{l=1}^{n} \mathrm{e}^{2\pi ijk_l 2^{-l}} |k_l\rangle && (2.8)\\
&= \frac{1}{2^{n/2}} \otimes_{l=1}^{n} \left[\sum_{k_l=0}^{1} \mathrm{e}^{2\pi ijk_l 2^{-l}} |k_l\rangle \right] && (2.9)\\
&= \frac{1}{2^{n/2}} \otimes_{l=1}^{n} \left[|0\rangle + \mathrm{e}^{2\pi ij2^{-l}} |1\rangle \right] && (2.10)\\
&= \frac{\left(|0\rangle + \mathrm{e}^{2\pi i 0.j_n} |1\rangle\right) \left(|0\rangle + \mathrm{e}^{2\pi i 0.j_{n-1}j_n} |1\rangle\right) \cdots \left(|0\rangle + \mathrm{e}^{2\pi i 0.j_1 j_2 \cdots j_n} |1\rangle\right)}{2^{n/2}}. && (2.11)
\end{aligned}
$$

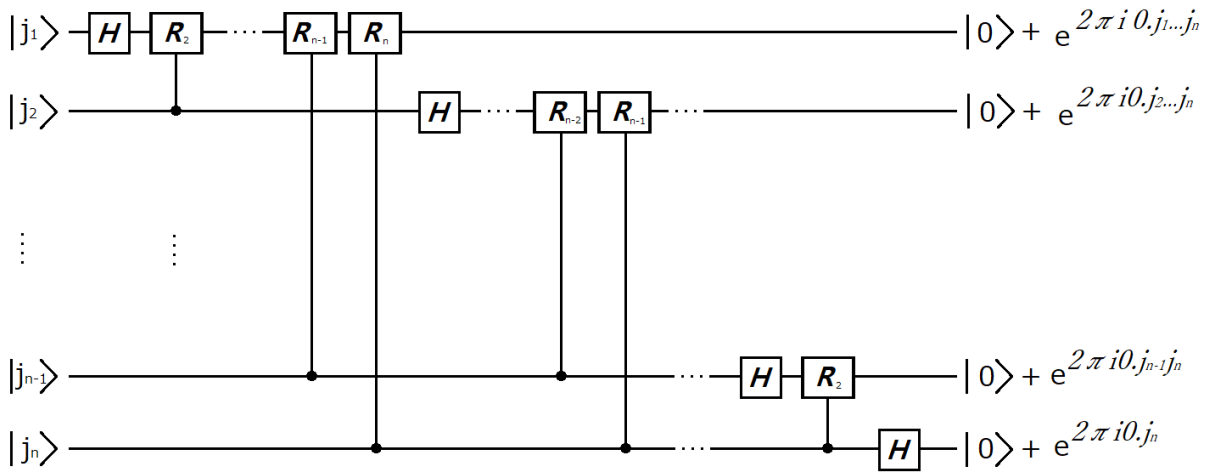


Fig. 2.1: The circuit of quantum Fourier transform

The circuit for quantum Fourier transform is shown in Figure 2.1. The gate R_k denotes the unitary transformation

$$R_k \equiv \begin{bmatrix} 1 & 0 \\ 0 & \mathrm{e}^{2\pi i/2^k} \end{bmatrix}. \tag{2.12}$$

To see that the circuit denotes the quantum Fourier transform, consider what happens in the input state $|j_1 \ldots j_n\rangle$. Applying the Hadamard gate to the first bit produces the state

$$\frac{1}{2^{1/2}} \left(|0\rangle + \mathrm{e}^{2\pi i 0.j_1} |1\rangle\right) |j_2 \ldots j_n\rangle, \tag{2.13}$$

since $e^{2\pi i 0.j_1} = -1$ when $j = 1$, and is $+1$ otherwise. Applying the controlled-R_2 gate produces the state U.

$$\frac{1}{2^{1/2}}\left(|0\rangle + e^{2\pi i 0.j_1 j_2}|1\rangle\right)|j_2 \dots j_n\rangle . \tag{2.14}$$

We continue applying the controlled-R_3, R_4 through R_n gates and have the state

$$\frac{1}{2^{1/2}}\left(|0\rangle + e^{2\pi i 0.j_1 j_2 \dots j_n}|1\rangle\right)|j_2 \dots j_n\rangle . \tag{2.15}$$

Next, we perform a similar procedure on the next qubit. The Hadamard gate puts us in the state and continue this procedure for each qubit. At the end we have the final state C

$$\frac{1}{2^{n/2}}\left(|0\rangle + e^{2\pi i 0.j_1 j_2 \dots j_n}|1\rangle\right)\left(|0\rangle + e^{2\pi i 0.j_2 \dots j_n}|1\rangle\right)\cdots\left(|0\rangle + e^{2\pi i 0.j_n}\right). \tag{2.16}$$

After the swap operation to reverse the order of the qubits, the state of the qubits is

$$\frac{1}{2^{n/2}}\left(|0\rangle + e^{2\pi i 0.j_n}\right)\left(|0\rangle + e^{2\pi i 0.j_{n-1} j_n}|1\rangle\right)\cdots\left(|0\rangle + e^{2\pi i 0.j_1 j_2 \dots j_n}|1\rangle\right). \tag{2.17}$$

2.3 Quantum phase estimation

Phase estimation is the key for many quantum algorithms and Fourier transform is the most important tool for Phase estimation[6]. Let operator U be unitary with eigenvector $|u\rangle$ correspond to eigenvalue $e^{2\pi i\varphi}$, where the value of φ is unknown. The goal of the phase estimation algorithm is to estimate φ. To estimate, we assume that we have available *blackboxes* (sometimes known as *oracles*) capable of preparing the state $|u\rangle$ and performing the controlled-U^{2^j} operation, for suitable non-negative integers j. The quantum phase estimation procedure has two registers. The first register contains t qubits in the initial state $|0\rangle$. the second register begins in the state $|u\rangle$ and has as many qubits as is necessary to store $|u\rangle$. Phase estimation is operated in several stages. First, we apply the circuit denoted in Figure 2.2. The starting of the circuit is applying a hadamard transform to the first register, followed by application of controlled-U operator on the second register. The state of the first register at the end is following formula

$$\frac{1}{2^{t/2}}\left(|0\rangle + e^{2\pi i 2^{t-1}\varphi}\right)\left(|0\rangle + e^{2\pi i 2^{t-1}}\right)\dots\left(|0\rangle + e^{2\pi i 2^0}|1\rangle\right) = \frac{1}{2^{t/2}}\sum_{k=0}^{2^t-1} e^{2\pi i\varphi k}|k\rangle . \tag{2.18}$$

The inverse quantum Fourier transform on the first register is key of the second stage of quantum phase estimation. This is obtained by reversing the circuit for the quantum Fourier

transform. The third and final stage of quantum phase estimation performs measurement to the state of the first register in the computational basis. To exactly express in t bits, as $\varphi = 0.\varphi_1 \dots \varphi_t$, the state R resulting from the first stage of phase estimation may be rewritten

$$\frac{1}{2^{t/2}} \left(|0\rangle + \mathrm{e}^{2\pi i 0.\varphi_t}\right) \left(|0\rangle + \mathrm{e}^{2\pi i 0.\varphi_{t-1}\varphi_t}\right) \cdots \left(|0\rangle + \mathrm{e}^{2\pi i 0.\varphi_1\varphi_2\cdots\varphi_t}\,|1\rangle\right). \tag{2.19}$$

A measurement in the computational basis therefore show us φ.

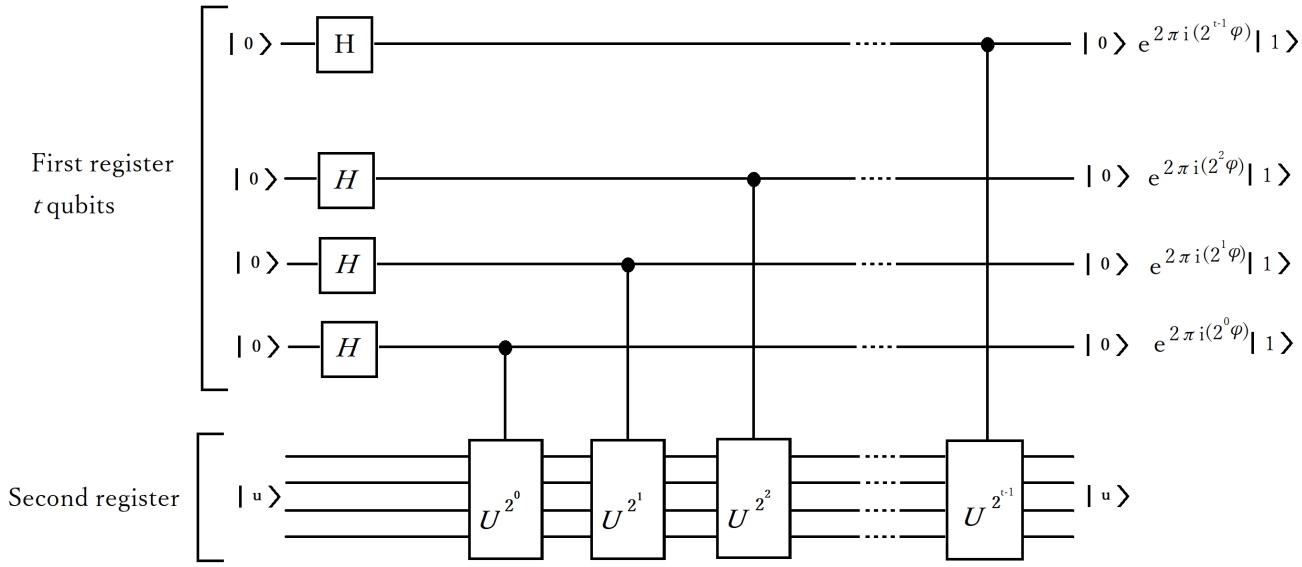


Fig. 2.2: The circuit of quantum phase estimation

Consequently, This algorithm allows us to estimate the phase φ representing an eigenvalue of a unitary operator U, given the corresponding eigenvector $|u\rangle$ by using the ability of the inverse Fourier transform to perform the transformation

$$\frac{1}{2^{t/2}} \sum_{j=0}^{2^t-1} \mathrm{e}^{2\pi i \varphi j}\,|j\rangle\,|u\rangle \to |\tilde{\varphi}\rangle\,|u\rangle\,, \tag{2.20}$$

where $\tilde{\varphi}$ is a approximation of φ when measured.

2.4 Hamiltonian simulation

One of the major potential application of quantum computer is the simulation of quantum system. The idea is proposed by Feynman in the concept of quantum computer. Most quantum algorithm for machine learning, and in particular the HHL algorithm, exploit quantum Hamiltonian simulation as a subroutine. Given a Hamiltonian operator $\hat{H}$, which is a Hermitian matrix, and the goal of Hamiltonian simulation is to implement a quantum circuit consising of the unitary operator $U = \mathrm{e}^{-iHt}$ for quantum state, up to given error, i.e. for quantum state $|\psi\rangle$, the evolution under a unitary operator is given by the time-independent Schrodinger equation:

$$i\frac{d}{dt}|\psi\rangle = \hat{H}|\psi\rangle, \tag{2.21}$$

The solution can be written as $|\psi(t)\rangle = \mathrm{e}^{-i\hat{H}t}|\psi\rangle$. In quantum case, as the number of qubits, the dimension of the Hilbert space increases exponentially, and thus any operator will be of exponential dimension. It makes that a quantum computer can be used to simulate the Hamiltonian operator, known as *Hamiltonian simulation*, which we wish to perform efficiently. The *efficient Hamiltonian simulation* is defined as follows.

2.4.1 Efficient Hamiltonian simulation

Definition 1 *We say that a Hamiltonian $\hat{H}$ that acts on n qubits can be efficiently simulated if for any $t > 0, e > 0$, there exists a quantum circuit $U_{\hat{H}}$ consisting of $poly(n, t, 1/\epsilon)$ gates such that $\|U_{\hat{H}} - \mathrm{e}^{-i\hat{H}t}\| < \epsilon$. Since any quantum computation can be implemented by a sequence of Hamiltonian simulations, simulating Hamiltonian in general is BQP-hard, where BQP refers to the complexity class of decision problems efficiently solvable on a universal quantum computer.*

Since it is NP-hard to find an approximate decomposition into elementary single- and two-qubit gates for a generic unitary and hence also for the evolution operator, simulating an arbitrary Hamiltonian efficiently is difficult except for certain classes of Hamiltonian having a particular structure. If H can be efficiently simulated, can simulate cH efficiently for any $c = poly(n)$, where n is the number of qubits, and e^{-iHt} is also simulated efficiently. Many natural Hamiltonian have the form of a sum of terms and then a sum of Hamiltonian is generalized by $Trotter - Suzuki\ methods$. Specifically, Graph-coloring method is applied

to *sparse matrix*. *sparse* and *sparse matrix* is defined as follows.

Definition 2 A N × N matrix is said to be *s-sparse* if it has at most s entries per row

Definition 3 A N × N matrix is said to be *sparse* if it has at most $poly(\log N)$ entries per row

2.4.2 Graph-coloring method

In the case that sparse matrix, we can simulate sparse Hamiltonian using Trotter-Suzuki method if given any index i, we can efficiently determine all of the indices j such as $\langle i|H|j\rangle \neq 0$, and moreover efficiently obtain the value of the corresponding elements of matrix, then we can simulate the Hamiltonian H efficiently. d means the maximum degree of graph. The problem is that if, given k colours, each edge is colored with restriction that there is no two same colored edge sharing a vertex. Vizing's theorem discuss the problem and conclude that for a graph with d, an edge coloring method exists with k at most $d + 1$.

Graph coloring method is based on graph theory. Let $G = (V, E)$ be an undirected graph where $|V| = N$, $|E| = M$, and A be adjacency matrix corresponding to G, where $A_{ij} = 1$ if the vertices i and j are connected, and $A_{ij} = 0$ otherwise. The first step is to find an edge-coloring of the graph associated with the sparse Hamiltonian H with k colors. Next, the graph can be decomposed by considering the subgraphs corresponding to each color. Thus, there exist decomposition of the original Hamiltonian in a sum of sparse Hamiltonians, containing at most $poly(\log |n|)$ terms. Indeed, adjacent matrix corresponding to a subgraph will be symmetric with at most one entry per row. Thus, each Hamiltonian term can be easily diagonalized and simulated.

One of the most important steps in this procedure is the classical algorithm to determine the edge coloring efficiently. however, it is possible to be required exponential time in case using Vizing's theorem to achieve decomposition, since finding an optimal coloring is an NP-complete problem. Thus, we should use other theorem to achieve the decomposition efficiently.

Lemma 1 *Suppose we are given an undirected, bipartite graph G with n vertices and maximum degree d, and that we can efficiently compute the neighbors of any given vertex. Then there is an efficiently computable edge coloring of G with at most colors.*

The proof of this lemma is easily explained to understand in [8]. Thus, using such useful

notations, we achieve to reduce the computational complexity of query to 1-sparse Hamiltonian requiring from $O(\log n)$ queries to $O(1)$ queries.

Chapter.3

METHOD OF STUDY

3.1 PRECEDENT STUDY

All previous studies have used the linearity of expected values and property of expectation to obtain lower bounds. In the studies, a variable is based on vertex.

3.1.1 Classical counterpart

Classical counterpart is the method for the computation of the complex Hamiltonian Schur form having a complexity of $O(\mathrm{n}^3)$ and being strongly backward stable. To discuss the process of algorithm, we will first transform the structured complex eigenvalue problem into a structured real problem of double size. After solving the eigenvalue problem for this extended matrix, we will recover the eigenvalue of the original matrix. i.e. for the Hamiltonian matrix, we partition N into $N_1 + iN_2$, real part and imaginary part. This partition is transformed into real skew-Hamiltonian Schur form through unitary matrices such as the Lie group of unitary symplectic matrices and permutation matrices. In real skew-Hamiltonian Schur form, the eigenvalues of the complex skew-Hamiltonian N can be computed by applying Van Loan's method is strongly backward stable it is clear that the computed eigenvalues of N are the exact eigenvalues of a real skew-Hamiltonian matrix. The detail of this algorithm is discussed in [1]. The total complexity of this classical algorithm is $O(\mathrm{n}^3)$.

3.1.2 Variational quantum estimation

The quantum phase estimation algorithm efficiently finds the eigenvalue of a given eigenvector in condition satisfying fully coherent evolution. However, the realization of the technique satisfying the condition seems a long way off in near future. On the contrary, variational Quantum estimation is hybrid quantum-classical algorithm for finding minimum eigenvalue of Hamiltonian and designed to take advantage of the strengths of a given architecture by replacing long coherent evolution with many short coherent evolutions.

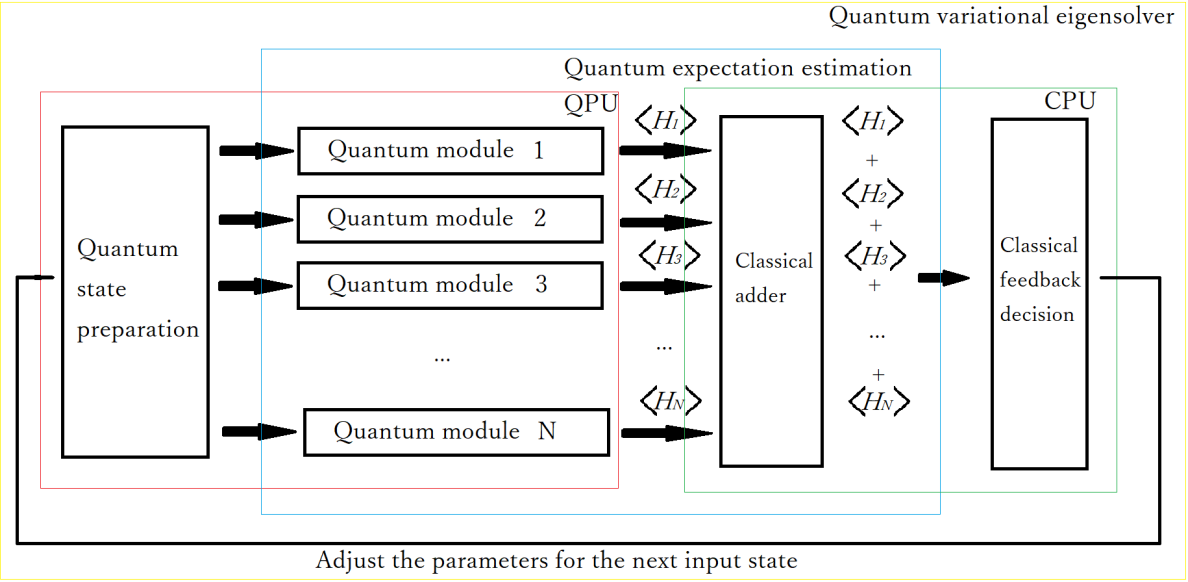


Fig. 3.1: Architecture of the quantum variational eigensolver

If some gates or quantum operations may be performed with higher fidelity, then the algorithm can leverage these strengths in the design of the quantum hardware ansatz. It is will known that the eigenvalue problem for an observable represented by an operator H can be restated as a variational problem on the Rayleigh-Ritz quotient, such that the eigenvector $|\psi\rangle$ corresponding to the lowest eigenvalue is the $|\psi\rangle$ that minimizes

$$\frac{\langle\psi| H |\psi\rangle}{\langle\psi|\psi\rangle} \tag{3.1}$$

By adjusting the parameters in preparation of $|\psi\rangle$ and computing the Rayleigh-Ritz quotient using estimation as a subroutine in a classical minimization, one may prepare unknown eigenvectors[4].

3.1.3 A quantum algorithm providing exponential speed increase for finding eigenvalues and eigenvectors

This study introduces a novel polynomial time quantum algorithm using quantum Fourier transform to find eigenvalues and eigenvector of Hamiltonian operator[9]. All classical counterparts require exponential time. The idea using quantum computer for the problem having specific structure is contrived from Feynman. It was based upon the observation that quantum systems are described in a Hilbert space whose size grows exponentially with the number of particles. The problem to be considered in that paper can be precisely stated

as follows. Given the time-evolution operator $U = \mathrm{e}^{-\frac{i}{\hbar}Ht}$ which corresponds to the Hamiltonian H, we find a true eigenvalue λ_v and true eigenvector V. In that paper, we assume that approximate eigenvector V_a of U that can be generated in quantum polynomial time is prepared as a priori knowledge, and a true eigenvalue λ_v is obtained by apply quantum phase estimation algorithm using quantum Fourier transform to approximate eigenvector V_a and operator U. It is observed with probability $\langle \mathrm{V}|\mathrm{V_a}\rangle^2$, since approximate eigenvector V_a is constructed by true eigenvalues of U. If the state V_a has the probability $\langle \mathrm{V}|\mathrm{V_a}\rangle^2$ not small, the approximate eigenvector V_a contains a component of the true eigenvector bounded by a polynomial function of the problem size. i.e. V and λ_v can be found in time proportional to $\frac{1}{\langle \mathrm{V_a}|\mathrm{V}\rangle^2}$ and $1/\epsilon$, where ϵ is the desired accuracy.

We will now sketch the algorithm for finding eigenvalue and eigenvector of given Hamiltonian through applying to any $\hat{U}$ that can be implemented in quantum polynomial time, corresponding to a given Hamiltonian. To begin, given a quantum computer consisting of $m + l + w$ qubits, where a total of m qubits are used quantum Fourier transform, a total of l qubits describe the Hilbert space in which the operator $\hat{U}$ acts, and w extra working qubits are required for temporary storage, Let $M = 2^m$. Assume that the m index qubits are initially in the state $|0\rangle$ and that the l qubits are initially in the state V_a. the initial state is as follows

$$|\psi\rangle = |0\rangle |V_a\rangle . \tag{3.2}$$

And then the Hadamard gate is applied to each qubit in m index qubits to obtain the state

$$|\psi\rangle = \frac{1}{\sqrt{M}} \sum_{j=0}^{M-1} |j\rangle |V_a\rangle \tag{3.3}$$

Next, a series of quantum logic operations that transform the state is performed to obtain the state

$$|\psi\rangle = \frac{1}{\sqrt{M}} \sum_{j=0}^{M-1} |j\rangle \left(\hat{\mathrm{U}}\right)^j |V_a\rangle . \tag{3.4}$$

At this point, label the eigenvalue of $\hat{U}$ by the state $|\phi_k\rangle$ and the corresponding eigenvalues with λ_k. the approximate eigenvalue V_a is rewritten as

$$|V_a\rangle = \sum_k c_k |\phi_k\rangle . \tag{3.5}$$

In the case that eigenvalue λ_k is form as $\mathrm{e}^{i\omega k}$, the state $|\psi\rangle$ is represented as

$$|\psi\rangle = \frac{1}{\sqrt{M}} \sum_{k} c_k \, |\phi_k\rangle \sum_{j=0}^{M-1} \mathrm{e}^{i\omega_k j} \, |j\rangle \, . \tag{3.6}$$

To obtain eigenvalues, one performs quantum Fourier transform to the state $|\psi\rangle$ and the eigenvalues λ_k is measured with the probability of $\langle \mathrm{V_a} | \phi_\mathrm{k} \rangle^2$. It means the more $\langle \mathrm{V_a} | \phi_\mathrm{k} \rangle^2$ is close to 1, the more the approximate eigenvector V_a is close to the true eigenvector.

In this method, it need priori knowledge of approximate eigenvector in advance and obtains the initial approximate eigenvector V_a by using classical algorithm. it has the possibilty occuring bottle neck in working classcial algorithm. and complexity of this method depend on similarity between the approximate and the true one. That's why it is difficult to draw the exact complexity of this algorithm.

Chapter.4

RESULT

4.1 Algorithm

We described a more detailed algorithm. Given $N \times N$ Hermitian matrix H with $N = 2^n$, n is the number of qubits, we want to transform a Hermitian matrix H into a unitary operator e^{iHt}. This is possible if H is s-sparse and efficiently row computable. It means matrix has at most s nonzero entries per row and for a row index these can be computed in time $O(s)$. In this conditions, [7] describe that how to simulate e^{iHt} in time

$$\tilde{O}(\log(N)s^2t), \tag{4.1}$$

where the $\tilde{O}(s)$ suppresses more slowly-growing terms. Let $|b\rangle$ be the state $|0\cdots00\rangle$ in n qubits. We also decompose $|b\rangle$ in the eigenvector basis of H, using quantum phase estimation. Let $|u_j\rangle$ be the eigenvectors of H, and λ_j be the corresponding eigenvalue of $|u_j\rangle$, and

$$|\psi_0\rangle := \sqrt{\frac{2}{T}} \sum_{\tau=0}^{T-1} \sin \frac{\pi(\tau + \frac{1}{2})}{T} |\tau\rangle \tag{4.2}$$

be the state of the first register for some large T.

Next we apply the conditional Hamiltonian evolution $\sum_{\tau=0}^{T-1} |\tau\rangle \langle\tau| \otimes e^{2\pi i H\tau/T}$ on $|\psi_0\rangle \otimes |b\rangle$. Quantum Fourier transform is applied to the first register to obtain the state

$$\sum_{j=1}^{N} \sum_{k=0}^{T-1} \alpha_{k|j}\beta_j |k\rangle |u_j\rangle , \tag{4.3}$$

where $|k\rangle$ are the Fourier basis states, and the more λ_j is close to k, the more $|\alpha_{k|j}|$ is large. We can also define $\tilde{\lambda_k} := k$ and rewrite the Fourier basis state to obtain

$$\sum_{j=1}^{N} \sum_{k=0}^{T-1} \alpha_{k|j}\beta_j |\tilde{\lambda_k}\rangle |u_j\rangle . \tag{4.4}$$

If the quantum phase estimation is performed perfectly, we would have $\alpha_{k|j} = 1$ if $\tilde{\lambda_k} = \lambda_j$, and 0 otherwise. Under this assumption, the state is as follows

$$\sum_{j=1}^{N} \beta_j \left|\tilde{\lambda_j}\right\rangle \left|u_j\right\rangle . \tag{4.5}$$

The eigenvalue $\tilde{\lambda_j}$ is measured by operator consisting of computational basis.

Up to now, we discuss the process of the algorithm in the case that the state $|b\rangle$ is $|0\cdots00\rangle$. The remainder of work is to apply this process to other state such that $|b\rangle$ is from $|0\cdots01\rangle$ to $|1\cdots1\rangle$. Each state from $|0\cdots01\rangle$ to $|1\cdots1\rangle$ consists of eigenvector of Unitary operator under the restriction that sum of square of each amplitude in each state is 1, and the component ratio is not the same. However, from the point of view of matrix representation of Unitary operator, each column/row is orthonormal and it makes that sum of square of amplitude for specific eigenvector $|u_j\rangle$ in all the state is 1. As the result, the probability of measuring the eigenvalue $|\tilde{\lambda_j}\rangle$ is uniform in the algorithm.

4.2 The runtime of the algorithm

To produce the state

$$|\psi_0\rangle = \sqrt{\frac{2}{T}} \sum_{\tau=0}^{T-1} \sin \frac{\pi\left(\tau + \frac{1}{2}\right)}{T} |\tau\rangle \tag{4.6}$$

for a T to be chosen later up to error ϵ_ψ in time poly $\log(T/\epsilon_\psi)$, we use the technique of [11].

Other process to implement our algorithm needs to Hamiltonian simulation. To simulate e^{iHt}, Hermitian H given, for some $t \geq 0$, we use the algorithm of [7]. If H is s-sparse, $t \leq 2\pi$ and we guarantee that the error is $\leq \epsilon_H$, then this requires time

$$\begin{aligned} T_H &= O(\log(N)(\log^*(\mathrm{N}))^2 s^2 9^{\sqrt{\log(s^2/\epsilon_H)}}) && (4.7)\\ &= \tilde{O}(\log(N)s^2) && (4.8) \end{aligned}$$

This process is repeated from $|0\cdots0\rangle$ to $|1\cdots1\rangle$ and becomes random sampling in consequence from the point of the view of eigenvalue $|\tilde{\lambda_j}\rangle$. The number of times to obtain all eigenvalue is the same answer of Coupon collector's problem and we should perform $N \log N$ times to obtain all the case. Thus, the total-runtime of our algorithm is

$$\tilde{O}(Ns^2(\log \mathrm{N})^2). \tag{4.9}$$

4.3 Error analysis

When applying normal quantum phase estimation to unitary operator given, it is successful with $\frac{4}{\pi^2}$ probability at least, and since the quantum phase estimation is based on quantum Fourier transform, the more value is close to the correct one, the larger probability to be measured is. Thus, it makes that even if you don't obtain the correct eigenvalue through the technique, the value measured is expected to be close to the correct than the value not using the technique. Common quantum phase estimation is using the initial state $|\psi\rangle = \frac{1}{T}\sum_{m=0}^{N}|m\rangle$. It has the merit such as having narrow width and high peak in terms of probability distribution. But it has slowly decreasing tails[11]. In HHL algorithm, it is solve by using other initial state $|\psi_0\rangle = \sqrt{\frac{2}{T}}\sum_{\tau=0}^{T-1}\sin\frac{\pi(\tau+\frac{1}{2})}{T}|\tau\rangle$ with probability distribution having a little bit wide and low but having tails decreasing much faster than $|\psi\rangle$. In this paper, we use the technique using the state $|\psi_0\rangle$ to obtain the stable value of eigenvalue and show the probability measuring correct value using the technique $|\psi_0\rangle$. $|\alpha_{k|j}|$ is as follows

$$
\begin{aligned}
\alpha_{k|j} &= \frac{\sqrt{2}}{T}\sum_{\tau=0}^{T-1}\mathrm{e}^{i\frac{\tau}{T}(2\pi\lambda_j-2\pi k)}\sin\frac{\pi(\tau+\frac{1}{2})}{T} && (4.10)\\
&= \frac{1}{i\sqrt{2}T}\sum_{\tau=0}^{T-1}\mathrm{e}^{i\frac{\tau\delta}{T}}\left(\mathrm{e}^{\frac{i\pi(\tau+1/2)}{T}}-\mathrm{e}^{-\frac{i\pi(\tau+1/2)}{T}}\right) && (4.11)\\
&= \frac{1}{i\sqrt{2}T}\sum_{\tau=0}^{T-1}\mathrm{e}^{\frac{i\pi}{2T}}\mathrm{e}^{i\tau\frac{\delta+\pi}{T}}-\mathrm{e}^{-\frac{i\pi}{2T}}\mathrm{e}^{i\tau\frac{\delta-\pi}{T}} && (4.12)\\
&= \frac{1}{i\sqrt{2}T}\left(\mathrm{e}^{\frac{i\pi}{2T}}\frac{1-\mathrm{e}^{i\pi+i\delta}}{1-\mathrm{e}^{i\frac{\delta+\pi}{T}}}-\mathrm{e}^{-\frac{i\pi}{2T}}\frac{1-\mathrm{e}^{i\pi+i\delta}}{1-\mathrm{e}^{i\frac{\delta-\pi}{T}}}\right) && (4.13)\\
&= \frac{1+\mathrm{e}^{i\delta}}{i\sqrt{2}T}\left(\frac{\mathrm{e}^{-i\delta/2T}}{\mathrm{e}^{-\frac{i}{2T}(\delta+\pi)}-\mathrm{e}^{\frac{i}{2T}(\delta+\pi)}}-\frac{\mathrm{e}^{-i\delta/2T}}{\mathrm{e}^{-\frac{i}{2T}(\delta-\pi)}-\mathrm{e}^{\frac{i}{2T}(\delta-\pi)}}\right) && (4.14)\\
&= \frac{(1+\mathrm{e}^{i\delta})\mathrm{e}^{-i\delta/2T}}{i\sqrt{2}T}\left(\frac{1}{-2i\sin(\frac{\delta+\pi}{2T})}-\frac{1}{-2i\sin(\frac{\delta-\pi}{2T})}\right) && (4.15)\\
&= -\mathrm{e}^{i\frac{\delta}{2}(1-\frac{1}{T})}\frac{\sqrt{2}\cos(\frac{\delta}{2})}{T}\left(\frac{1}{\sin(\frac{\delta+\pi}{2T})}-\frac{1}{\sin(\frac{\delta-\pi}{2T})}\right) && (4.16)\\
&= -\mathrm{e}^{i\frac{\delta}{2}(1-\frac{1}{T})}\frac{\sqrt{2}\cos(\frac{\delta}{2})}{T}\cdot\frac{\sin(\frac{\delta-\pi}{2T})-\sin(\frac{\delta+\pi}{2T})}{\sin(\frac{\delta+\pi}{2T})\sin(\frac{\delta-\pi}{2T})} && (4.17)\\
&= \mathrm{e}^{i\frac{\delta}{2}(1-\frac{1}{T})}\frac{\sqrt{2}\cos(\frac{\delta}{2})}{T}\cdot\frac{2\cos(\frac{\delta}{2T})\sin(\frac{\pi}{2T})}{\sin(\frac{\delta+\pi}{2T})\sin(\frac{\delta-\pi}{2T})} && (4.18)
\end{aligned}
$$

We consider the case that the probability of measuring the correct value only allowing error by quantum Fourier transform and ignore phase. Thus, the equation has lower bound

as follows

$$> \frac{2\sqrt{2}\left(1-\frac{2}{\pi}|\frac{\delta}{2}|\right)\left(1-\frac{2}{\pi}|\frac{\delta}{2T}|\right)\left(\frac{2\pi}{\pi 2T}\right)}{T\left(\frac{\delta+\pi}{2T}\right)\left(\frac{\delta-\pi}{2T}\right)} \tag{4.19}$$

$$= \frac{8\sqrt{2}\left(1-\frac{1}{\pi}|\delta|\right)\left(1-\frac{1}{\pi}|\frac{\delta}{T}|\right)}{(\delta^2-\pi^2)} \tag{4.20}$$

$$= \frac{8\sqrt{2}\left(\pi-|\delta|\right)\left(\pi-|\frac{\delta}{T}|\right)}{\pi^2\left(\delta+\pi\right)\left(\delta-\pi\right)}. \tag{4.21}$$

From the error caused by quantum Fourier transform, and ignoring phases, then the lower bound of $|\alpha_{k|j}|$ is derived:

$$|\alpha_{k|j}| > \frac{4\sqrt{2}}{\pi^2}(1-\frac{1}{T}). \tag{4.22}$$

Chapter.5

CONCLUSION

5.1 REMARKS

Many studies using quantum algorithm have used a priori knowledge from preparing approximate eigenvector and been focused on obtaining specific eigenvector and eigenvalue such as ground state and energy until now. Since they should have process through preparing approximate eigenvector that they do not know how the approximate is close to the correct accurately, It is difficult to obtain the accuracy runtime, not the possibility.

In this study, using the properties of unitary operator such that row/column of unitary operator is orthonormal, we devise the algorithm not using priori knowledge such as approximate eigenvector to obtain all eigenvalues. Our algorithm uses the technique using the initial state $|\psi_0\rangle$ to measure the eigenvalue corresponding to eigenvector stably and has steps such as using Hamiltonian simulation, quantum phase estimation during $\tilde{O}(s^2 \log N)$ run time. These steps is applied with the initial states set, from $|0\cdots0\rangle$ to $|1\cdots1\rangle$, and applying to the set is repeated $\log N$ times to measure all eigenvalue of matrix given. Thus, On the contrary that the classical algorithm to find all eigenvalue of Hermitian matrix given is performed during $O(\mathrm{N}^3)$ runtime, the total runtime of our novel algorithm becomes $\tilde{O}(Ns^2(\log \mathrm{N})^2)$.

Our quantum algorithm using quantum phase estimation exploiting the initial state used in HHL algorithm has distribution of the eigenvalue, having slightly wider and lower central peak than normal QPE, but having rapidly decreasing tails[11]. Thus, probability measuring the correct eigenvalue allowing error by quantum Fourier transform exceeds $\frac{32}{\pi^4}(1-\frac{1}{T})^2$.

5.2 FUTURE RESEARCH

Our algorithm is used in the condition that given matrix is Hermitian or Unitary. It is useful to obtain eigenvalues of Hamiltonian. however, our algorithm is applied to limited case such as Hermitian and Unitary. Finding eigenvalue of matrix is one of the important problems in mathematics, science and engineering field. Thus, next our study is that

designing quantum algorithm for finding all eigenvalues of more general matrix than Hermitian matrix, i.e. we generalize our quantum algorithm for more general matrix. Quantum phase estimation is composed with several techniques of quantum algorithm. Among the techniques, quantum Fourier transform is the most important one. However, improved quantum phase estimation based on quantum walk, not based on quantum Fourier transform, appear lately[12]. It is more accurate than traditional method. Thus, if our algorithm based on the improved, then it is expected to bring lots of room for improvement to our algorithm.

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